

Team 30: Object Recognition in Refrigerator Scenario

Kaixin Zhang
7076400

Vaishnavi Borse
7075846

1. Abstract

We study open-world ingredient recognition in refrigerator scenes using a hybrid pipeline that combines YOLO for proposals, SAM for mask-tightening, BLIP for text priors, and CLIP for semantic re-ranking. On our test set, the class-agnostic mAP improves from 0.069 to 0.083 after SAM refinement (+20% relative). Under class-aware evaluation—our hardest setting—the final pipeline lifts mAP@0.5:0.75 from 0.0034 to 0.0260 (7.7×), while discovering categories outside the YOLO training label set (e.g., spray). We analyze where gains come from, the precision–recall trade-offs, and failure modes (container bias, visually similar dairy).

2. Introduction

Detecting fine-grained fridge items is challenging due to small object sizes, heavy occlusion, and visually similar packaging (e.g., milk vs. yogurt vs. cheese). Supervised detectors often miss tail classes or overfit to textures. We ask: can we leverage vision-language priors to re-score detections and inject scene priors, without re-training the detector? Our contributions: A training-free hybrid pipeline: YOLO proposals → SAM box tightening → BLIP allow-list → CLIP re-scoring/relabeling with class-specific gates. A name canonicalization step (aliases → canonical classes) that de-duplicates noisy labels. A detailed analysis of per-class precision/recall and error modes, showing consistent gains on several difficult categories.

3. Related Work

- YOLO-family models are strong one-stage detectors but can struggle with long-tail labels and small objects.
- SAM provides high-quality masks for box refinement.
- CLIP aligns image–text and is widely used for zero-/few-shot classification and open-vocabulary detection.
- BLIP produces diverse captions useful as image priors.

- Recent works combine detector proposals with CLIP-style re-labeling and prompt engineering; we instantiate a practical version tailored to fridge items with per-class gates and scene allow-lists.

4. Method

4.1. Canonical class space

We unify noisy labels via an alias map (e.g., apples→apple, paper box/plastic box→box, olives/olive→olive). Duplicated categories in the GT are merged in-memory by remapping category_id to a primary id; COCO indices are rebuilt. This reduces class fragmentation and confusion.

4.2. Proposal generation and box refinement

- YOLO (Ultralytics) yields candidate boxes. We optionally apply flip TTA and consolidate with class-agnostic WBF + NMS.
- SAM refine. For each box (xyxy), SAM predicts masks; we convert the top-score mask to a tighter xywh box. A gate accepts the refined box only if IoU(old, new) 0.60 and area ratio [0.30, 1.10], preventing harmful drifts.

4.3. Scene allow-list with BLIP

We generate Top-12 diverse captions per image (nucleus sampling), parse noun chunks with spaCy, normalize tokens, and map to canonical class names. This builds an image-level allow-list of plausible classes. Predictions outside the allow-list are soft-down-weighted (not hard-rejected), which helps recall.

4.4. CLIP re-scoring and re-labeling

For each refined crop, we compute CLIP similarity against all classes and take the top class with score s_{clip} (normalized to [0,1]). We fuse with the detector confidence s_{yolo} by $s_{fused} = s_{clip} * s_{yolo}$. We apply:

- Semantic gate: $s_{clip} \geq \tau_{clip}(c)$ (higher for high-FP classes like milk, butter, cheese, bread, box, bottle).

- Fused gate: $s_{fused} \geq \tau_{fused}(c)$ (stricter for high-FPs).
- Soft salvage: if s_{yolo} is low but s_{clip} is very strong, keep with reduced threshold.
- Per-image per-class Top-K (K=2) after class-wise NMS (IoU 0.65) to limit duplicates.

5. Experimental Results and Analyses

5.1. Experimental Setup

- Data. 33 test images; YOLO trained on 3000 images for 20 epochs (class names cleaned and merged).
- Metrics. COCO AP/AR. We report both class-agnostic (useCats=0) and class-aware (useCats=1).
- Key thresholds. MIN_CONF_SOFT=0.05, CLIP_TAU_DEFAULT=0.32, CLIP_TAU_HIGHFP=0.40, FUSED_TAU=0.06, FUSED_TAU_HIGHFP=0.09, class-wise NMS IoU=0.65, Top-K=2.

5.2. Experimental Results and Analyses

5.2.1 Class-agnostic localization (useCats=0)

Method	mAP@[.5:.95]	mAP@0.5	mAP@0.75	AR@1	AR@10	AR@100
YOLO baseline	0.0691	0.2051	0.0382	0.020	0.138	0.246
YOLO + SAM	0.0830	0.2327	0.0431	0.026	0.148	0.268

Figure 1. Ablation bar chart (mAP@[.5:.95], @0.5, @0.75, AR@1, @10, @100) comparing YOLO vs YOLO+SAM..

Takeaway. SAM-tightening improves localization quality across thresholds (+20% relative @ mAP@[.5:.95]), suggesting tighter boxes help bigger objects.

5.2.2 Class-aware detection (useCats=1)

Method/Stage	mAP@[.5:.95]	mAP@0.5	mAP@0.75
YOLO(class-aware)	0.0034	0.0069	0.0029
Full hybrid(ours)	0.0260	0.0530	0.0260

Figure 2. bar chart (mAP@[.5:.95], @0.5, @0.75,) comparing YOLO baseline vs ours.

Relative gain: 7.7× at mAP@0.5 and 9× at mAP@[.5:.95] over YOLO (class-aware).

class	precision	recall	TP	FP	FN
pickle	0.500	0.750	3.000	3.000	1.000
yogurt	0.462	0.500	6.000	7.000	6.000
tomato	0.417	0.556	5.000	7.000	4.000
jam	0.400	0.750	6.000	9.000	2.000
cheese	0.333	0.083	1.000	2.000	11.000
milk	0.231	0.375	3.000	10.000	5.000
carrot	0.125	0.091	2.000	14.000	20.000
medicine	0.100	0.500	1.000	9.000	1.000
spray	0.083	0.500	1.000	11.000	1.000
sauce	0.056	0.500	1.000	17.000	1.000

Figure 3. Per-class P/R (top 10 classes).

5.2.3 Error Analysis

- Why class-agnostic looks much better: localization quality is okay, but class assignment is the bottleneck.
- Scene priors help: BLIP allow-list reduces implausible classes, but we keep a soft penalty to avoid over-filtering rare but real items.
- Semantic gates vs. recall: Raising CLIP_TAU/FUSED_TAU sharply reduces FPs on milk/butter/box, but hurts recall on small/occluded items.
- Small objects: AP(small) is reported as 1.000 (no small-area instances under COCO’s definition in this split), so conclusions are only for medium/large in this test set.

6. Conclusion

We presented a training-free hybrid pipeline that uses SAM for box refinement, BLIP for scene priors, and CLIP for semantic re-scoring. On a small but challenging fridge dataset, it substantially improves class-aware detection over a YOLO baseline (mAP@[.5:.95] 0.003 → 0.026; mAP@0.5 0.007 → 0.053), and recovers categories that YOLO struggles with (e.g., spray, yogurt, pickle, tomato, jam). The main bottleneck is recall; future work will (i) add proposal diversification and region-wise BLIP, (ii) refine prompts/aliases per confusion group (milk/yogurt/cheese/butter), and (iii) learn lightweight calibrations for class thresholds.

7. References

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